APFEA

APFEA Reports

DemoSoC-1 — Foundry Process Agent, Pre-Tape-Out Margin, and Control Tower reports combined in one document.

AI-assisted Phenomenological Foundry Engineering Approach (APFEA). All data in these reports are the synthetic DemoSoC-1 sample data (DEMO-PDK, 3 nm-class values for demonstration only). The reports are planner-level: they observe, attribute, and recommend — they do not run circuit simulation and do not actuate fab equipment.

How the three reports fit together

- Pre-Tape-Out Margin Report (Part One) — the pre-tape-out check: design against anticipated VTs (VTs), not VT0.
- Foundry Process Agent Report — the full-flow judgement across design, tapeout, fab, interposer, wafer sort, packaging, and final test.
- Control Tower Report (Part Two) — the silicon closed loop: realized shifts refit the rules for the next tape-out.

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1 — Foundry Process Agent Report

DemoSoC-1 (synthetic sample customer chip)

- **Main chip:** SoC ASIC
- **Stages observed:** design, tapeout, fab, interposer, wafer_sort, packaging_3d, final_test

Stage results

- **design:** DOE PASS
- **fab:** VT REFIT APPLIED
- **packaging_3d:** MULTI-DIE GATE: main SoC ASIC + DRAM, HBM, photonics, analog

Block health

Block	Health
CPU cluster	GREEN
GPU / fabric	GREEN
IO bank	GREEN
SRAM cache	RED
PLL	GREEN
SerDes 112G	GREEN

Actions

- HOLD SPEC: SRAM cache is foundry-fixed and stays at VT0; remedy is CMOS-side only (LVT center within +/-4.0% of VT0)
- Apply silicon-refit shift rules to the next tape-out (est. error 0.50% -> 0.26%)

2 — APFEA Pre-Tape-Out Margin Report

Prototype v0.2

- **PDK:** DEMO-PDK (synthetic, 3nm-class values for demonstration only)
- **Customer chip:** DemoSoC-1 (synthetic sample customer chip) — type soc_large, die 520 mm², specialty devices: serdes, pll
- **Shift rules apply:** yes (chip type gate in config/shift_rules.json)
- **Residual estimation error carried:** 0.50%
- **Business model:** foundry — foundry-fixed blocks (SRAM) keep VT0, spec untouchable; only CMOS VTs change

Step I — Anticipated VTs

CMOS designs against VTs, not VT0.

Flavor	Pol	VT0 (V)	Shift	VTs (V)	sigma (% of |VT|)
HVT	N	+0.420	+0.0%	+0.4200	2.0
HVT	P	-0.400	+0.0%	-0.4000	2.0
LVT	N	+0.280	-6.0%	+0.2632	2.3
LVT	P	-0.260	-6.0%	-0.2444	2.3
RVT	N	+0.350	+2.0%	+0.3570	2.2
RVT	P	-0.330	+2.0%	-0.3366	2.2
ULVT	N	+0.220	-4.0%	+0.2112	2.2
ULVT	P	-0.200	-4.0%	-0.1920	2.2

Foundry-fixed blocks (e.g. SRAM) are NOT shifted: they keep VT0 as their design point, exactly as the foundry supplies them. VTs apply to the customer's CMOS design only.

Silicon corner splits around VTs

|VT|, N devices, CMOS flavors.

Flavor	-3sigma	-2sigma	-1sigma	+0sigma	+1sigma	+2sigma	+3sigma
HVT	0.3948	0.4032	0.4116	0.4200	0.4284	0.4368	0.4452
LVT	0.2450	0.2511	0.2571	0.2632	0.2693	0.2753	0.2814
RVT	0.3334	0.3413	0.3491	0.3570	0.3649	0.3727	0.3806
ULVT	0.1973	0.2019	0.2066	0.2112	0.2158	0.2205	0.2251

Step II — Block checks (continued)

Customer CMOS blocks (worst-case deviation vs block window)

Block	Flavor	Window	Conv. flow dev	Conv.	APFEA flow dev	APFEA
CPU cluster	RVT	9.0%	8.6%	PASS	7.1%	PASS
GPU / fabric	LVT	8.0%	12.9%	**AT RISK**	7.4%	PASS
SerDes 112G	ULVT	7.5%	10.6%	**AT RISK**	7.1%	PASS
PLL	RVT	8.0%	8.6%	**AT RISK**	7.1%	PASS
IO bank	HVT	10.0%	6.0%	PASS	6.5%	PASS

Conventional flow designs against VT0, so silicon deviates from the design center by |shift| + 3 sigma in the worst case. The APFEA flow designs against VTs, leaving only 3 sigma + residual estimation error.

Foundry-fixed blocks (design point stays at VT0 — spec untouchable)

Block	Flavor	Center tolerance	Anticipated center shift	Check
SRAM cache	LVT	+/-4.0%	-6.0%	**AT RISK**

The conventional flow never performs this check before tape-out; APFEA performs it in Step II. A violated constraint is remedied on the CMOS side only — design centering, flavor strategy, process targeting with the foundry. The fixed block's spec is never an action item.

Verdict

- CMOS blocks at risk under the conventional (VT0) flow: **3 / 5** — each is a mask-fix re-tape-out exposure.
- CMOS blocks at risk under the APFEA (VTs) flow: **0 / 5**.
- **Foundry-fixed constraint violated**: SRAM cache (LVT) — anticipated center shift -6.0% exceeds its +/-4.0% tolerance around VT0. HOLD THE SPEC: SRAM cache stays at VT0; pull the LVT center back via CMOS-side centering and foundry process targeting.
- **VTs1 for silicon corner lots**: VTs (anticipated true center from step I). Use VTs as the center for real silicon corner-lot splits, subject to the foundry-fixed block constraints: the remedy for a violated constraint is CMOS-side (design centering, flavor strategy, process targeting with the foundry) — the fixed block's spec is never touched.

Scope note: planner-level margin bookkeeping per APFEA v3 Part One. No circuit simulation is run; block windows, tolerances and shift rules are configuration to be fitted from each foundry/design-house pair's own history.

3 — APFEA Control Tower Report

Prototype v0.2

- **Program:** DemoSoC-1 rev A (first silicon)
- **Current stage reached:** Final test
- **Program health:** **RED**

Flow status

- **Design:** done
- **Tape-out:** done
- **Fabrication:** done
- **Interposer:** done
- **Wafer sort:** done
- **2.5D/3D packaging:** done
- **Final test:** done
- **PCBA:** pending
- **System:** pending

VT feedback — anticipated vs realized

Fab corner-lot readout.

Flavor	Anticipated shift	Realized shift	Delta (estimator bias)
HVT	+0.00%	+0.31%	+0.31%
LVT	-6.00%	-5.29%	+0.71%
RVT	+2.00%	+2.40%	+0.40%
ULVT	-4.00%	-4.59%	-0.59%

RMS estimator bias this cycle: 0.53%. Residual estimation error for the next tape-out: 0.50% -> 0.26% (refit from silicon, floored — calibration shrinks humility, never erases it).

Refit shift rules for the next tape-out (config/shift_rules.json values):

- **HVT:** +0.31%
- **LVT:** -5.29%
- **RVT:** +2.40%
- **ULVT:** -4.59%

Rev-B margin re-check

Refit rules, Part One DOE re-run.

Block	Flavor	Window	Worst-case dev	Verdict
CPU cluster	RVT	9.0%	6.9%	PASS
GPU / fabric	LVT	8.0%	7.2%	PASS
SRAM cache (foundry-fixed @VT0 — center check)	LVT	4.0%	5.3%	**AT RISK**
SerDes 112G	ULVT	7.5%	6.9%	PASS
PLL	RVT	8.0%	6.9%	PASS
IO bank	HVT	10.0%	6.3%	PASS

Block health (downstream data)

- **CPU cluster:** GREEN — wafer-sort yield 99.2%
- **GPU / fabric:** GREEN — wafer-sort yield 98.7%
- **IO bank:** GREEN — wafer-sort yield 99.8%
- **PLL:** GREEN — final test: pass all corners
- **SRAM cache:** RED — wafer-sort yield 91.5%
- **SerDes 112G:** GREEN — final test: pass all corners; eye margin nominal
- **Note:** SRAM fails cluster on low-|VT| corner lots — consistent with the Part One redesign flag.
- **Note:** Timing-sensitive blocks validate the VTs-centered design margins.

Open actions

- HOLD SPEC: SRAM cache is foundry-fixed and stays at VT0 — silicon yield loss CONFIRMED; remedy is CMOS-side only: pull the LVT silicon center within +/-4.0% of VT0 via CMOS VTs centering and foundry process targeting. Never redesign the fixed block.
- Apply silicon-refit shift rules to the next tape-out (est. error 0.50% -> 0.26%)

Scope note: the tower observes, attributes, and recommends; it does not actuate fab equipment. Planner-level bookkeeping per APFEA v3 Part Two.